Evaluating Acoustic Keystroke Recognition under Random and Adversarial Noise

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https://github.com/NPM26103/Research-Methodology.git

1 Introduction & Motivation

This project investigates how added noise affects the performance of a deep learning model designed to identify keyboard keys from their acoustic signatures. The research is inspired by the foundational work of Harrison et al. (2023) [1], titled *A Practical Deep Learning-Based Acoustic Side Channel Attack on Keyboards*, which demonstrated the feasibility of acoustic keystroke recognition using recordings captured from ordinary devices.

2 Research Question & Hypothesis

- **Research Question:** At the same signal-to-noise ratio (SNR), does gradient-based adversarial noise reduce keystroke recognition accuracy more than random Gaussian noise?
- **Hypothesis:** SNR-calibrated Fast Gradient Sign Method (FGSM-sign) perturbations will cause a significantly greater reduction in keystroke recognition accuracy than random Gaussian noise, with **20 dB** serving as the primary benchmark for comparison.

3 Methodology

3.1 Dataset & Preprocessing

We will utilize the publicly available "Phone" dataset, which covers 36 alphanumeric keys. Audio recordings will undergo the following preprocessing pipeline:

1. Segmentation into individual keystrokes.
2. Manual verification and cleaning of audio segments.
3. Stratified splitting into training (60%), validation (20%), and test (20%) sets.

3.2 Model Architecture & Training

We employ a CoAtNet-based classifier adapted from public implementations [2] using PyTorch and the `timm` library.

- **Input Feature:** Log-Mel spectrograms extracted from audio segments.
- **Model Selection:** The architecture will be trained on the training split, and hyperparameter selection/checkpointing will be guided by performance on the validation split.

3.3 Experimental Setup & Perturbations

The trained model will be frozen (fixed weights) for evaluation under three environmental conditions:

1. **Clean Baseline:** Original test audio without added noise.
2. **Random Noise:** Additive White Gaussian Noise (AWGN) across 3 random seeds.
3. **Adversarial Noise:** White-box FGSM-sign perturbations.

Noise levels will be calibrated to exact SNR targets of **30 dB**, **20 dB**, and **10 dB**, verified by measuring actual SNR after signal clipping.

4 Evaluation Metrics & Statistical Analysis

Model performance across all conditions will be evaluated using:

- Overall Accuracy and Macro-F1 Score.
- Absolute and Relative Accuracy Reduction.
- Attack Success Rate (ASR) computed over initially correct predictions.
- Paired bootstrap confidence intervals to evaluate statistical significance between Gaussian and FGSM degradation.

5 Expected Contributions & Limitations

- **Contribution:** A fully reproducible, empirical assessment comparing model sensitivity to non-targeted random versus white-box adversarial acoustic perturbations.
- **Limitations:** The scope of the conclusions is restricted to digital, white-box attack scenarios within the controlled environment of the benchmark dataset.

References

[1] Harrison, J., Toreini, E., & Mehrnezhad, M. (2023). *A Practical Deep Learning-Based Acoustic Side Channel Attack on Keyboards*. arXiv preprint arXiv:2308.01074. https://arxiv.org/pdf/2308.01074

[2] Milutinovic, M. (2023). *Keyboard Acoustic Side-Channel Attack using CoAtNet*. GitHub repository. https://github.com/MomirMilutinovic/keyboard-acoustic-side-channel-attack-coatnet.git